VICTORY 2027

PHOTOSYNTHESIS
(PHOTOSYNTHESIS PROCESS)

LECTURE-02 BIOLOGY

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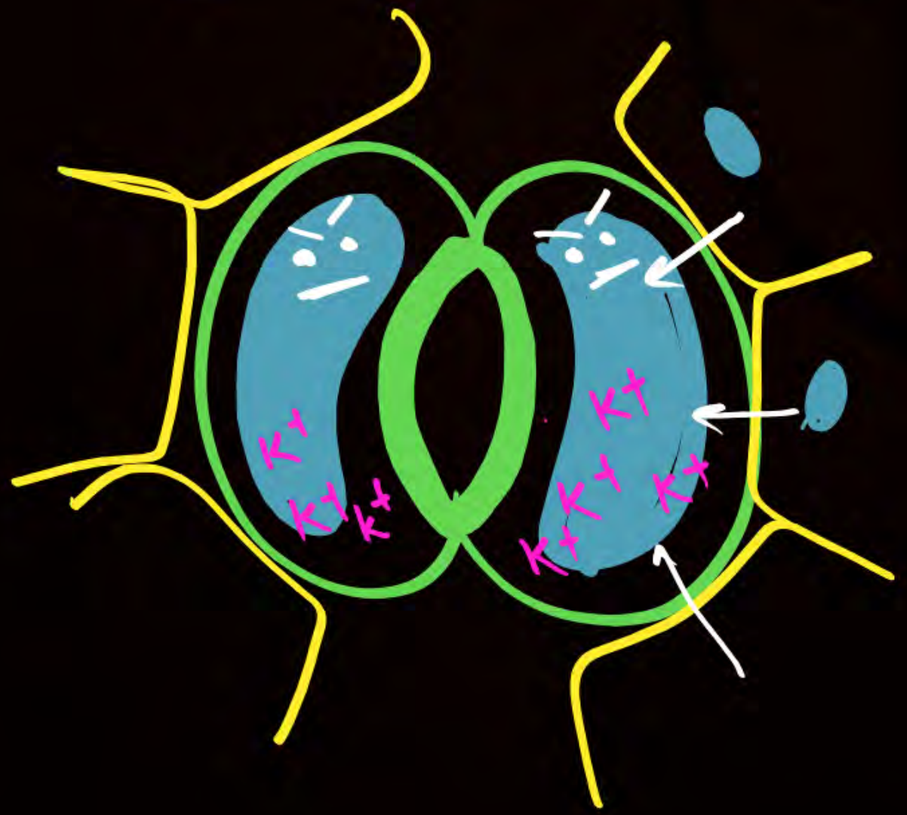
TOPICS TO BE COVERED

→ Sugar concentration theory

→ Light and Dark Reaction.


HELLO
NINJAS
忍者参上
影の中で戦う
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忍術最強
忍者
忍者参上

[K+ ion concentration]



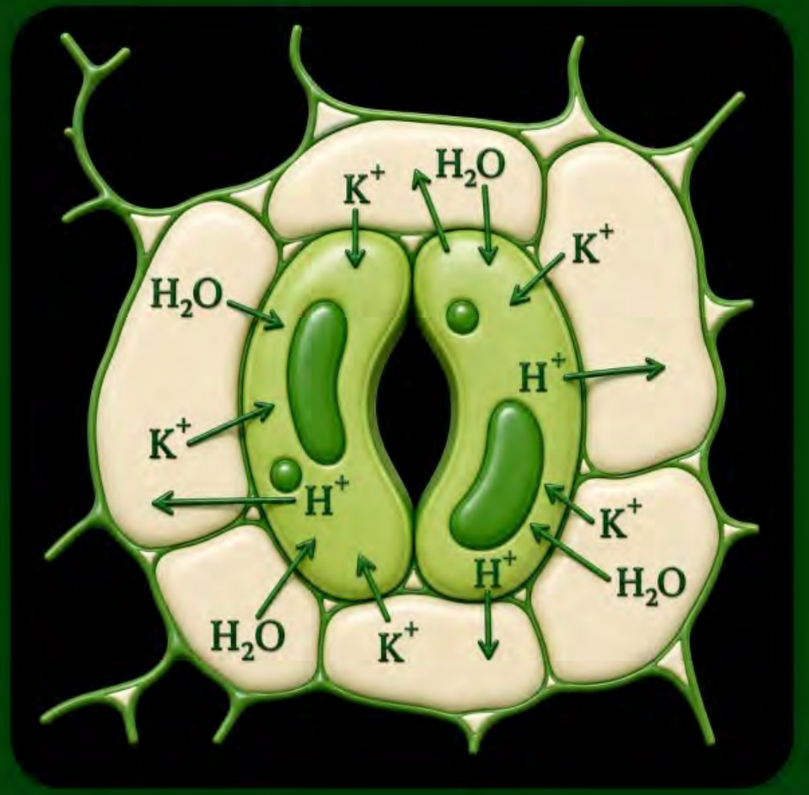
A) Active

B) Passive

K^+ Ion Concentration Theory

Opening of Stomata

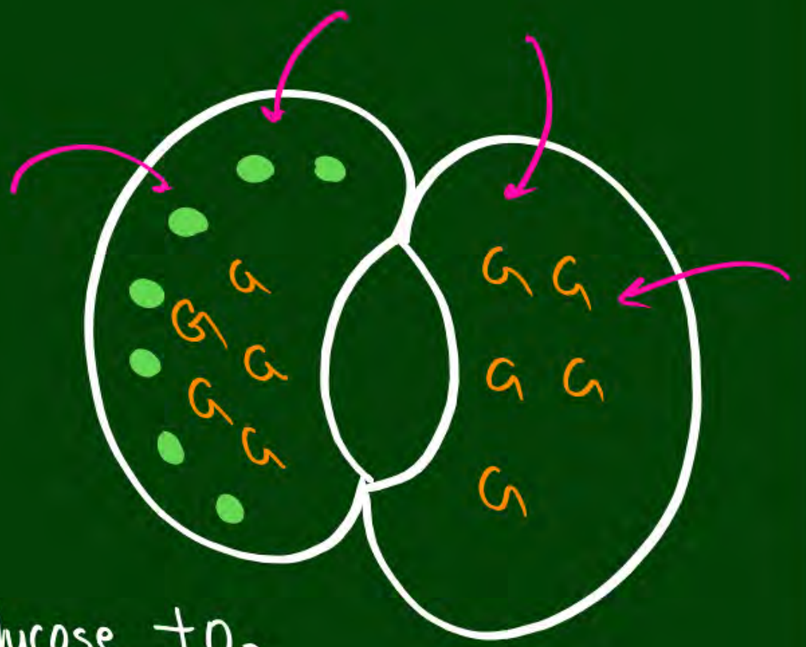
- Guard cells uses ATP actively to pump the potassium ions of epidermal cells
- Due to the high K^+ concentration, water enters the guard cells by osmosis.
- The guard cells become turgid (swollen).The swollen guard cells bend outward, and the stomata open.



Sugar Concentration Theory

- During daytime, the guard cells begin photosynthesis and the sugar (glucose) produced during the process increases the osmotic pressure which draws in water from the adjoining cells due to endosmosis. Hence, the guard cells become turgid and bulge outwards.

(Glucose)



$$CO_2 + H_2O \xrightarrow[\text{Chlorophyll}]{\text{sunlight}} \text{Glucose} + O_2$$

QUESTION

(vii) During sunlight, the guard cell turn flaccid to open the stomata.

1. True
2. False

Process of Photosynthesis

$CO_2 + H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} \text{Glucose} + O_2$

- CO_2 → from stomata
- H_2O → Absorbed by roots
- Chlorophyll → present in green parts of plant

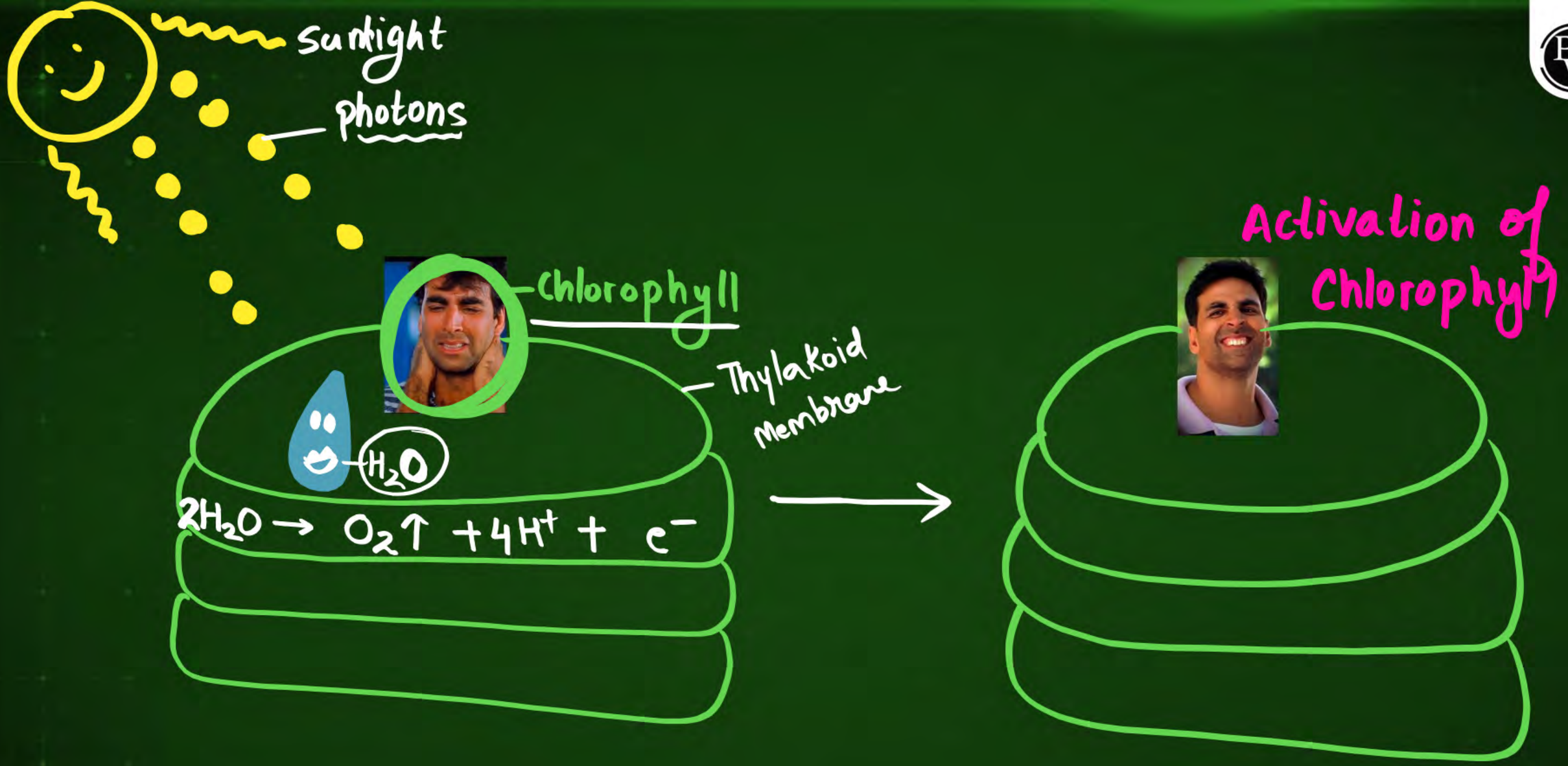
Phases of Photosynthesis

- Carbon dioxide from the atmosphere enters the leaf by diffusion through the stomata.
- Water from the soil is taken up by the roots, sent up through the stem and finally to the leaves where it is distributed in the mesophyll tissue.

$$6CO_2 + 12H_2O \xrightarrow[\text{Light energy}]{\text{Chlorophyll}} C_6H_{12}O_6 + 6H_2O + 6O_2 \uparrow$$

Light-dependent Phase (Photochemical Phase

- This phase is dependent on light and therefore, the phase is called the photochemical phase.
- The light reaction takes place in thylakoids (containing chlorophyll) of the chloroplasts.


sunlight
photons
Chlorophyll
Thylakoid Membrane
H_2O
$2H_2O \rightarrow O_2\uparrow + 4H^+ + e^-$
Activation of Chlorophyll



$$NADP + H^+ \rightarrow NADPH$$



NADP — Nicotinamide Adenine Dicnucleotide

ADP — Phosphate



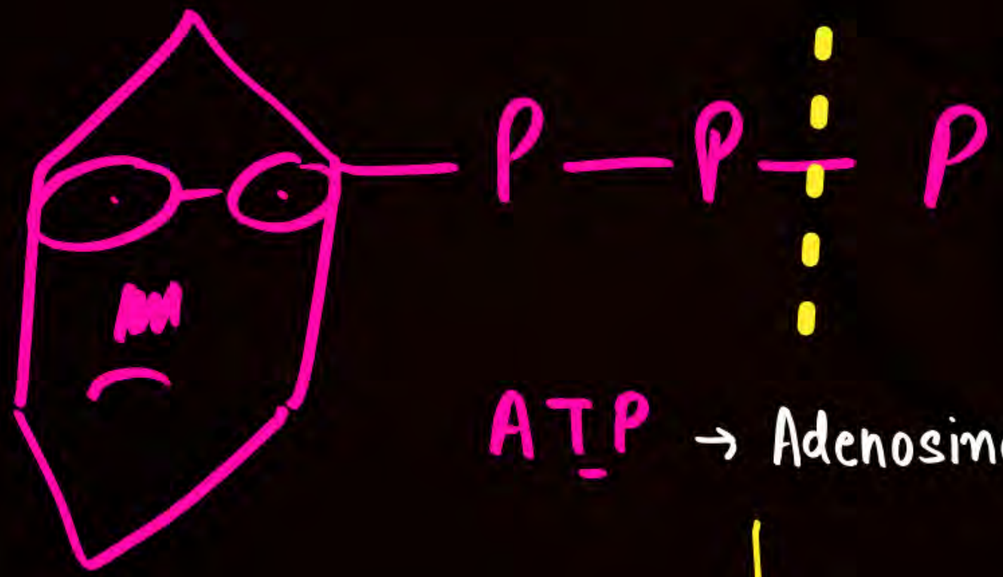
NADPH — Hydrogen Phosphate

$$\underset{2}{\underline{ADP}} + 4e^- + \underline{ip} \rightarrow ATP$$

?

ATP → ADP → AMP

AMP + ip
↓
ADP + ip
↓
ATP

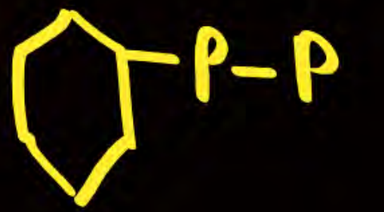


ATP → Adenosine Tri Phosphate (full energy)

↓

Adenine

ADP → Adenosine Di Phosphate



AMP → Adenosine Mono Phosphate

- **The light reaction (photochemical phase) occurs in two main steps.**

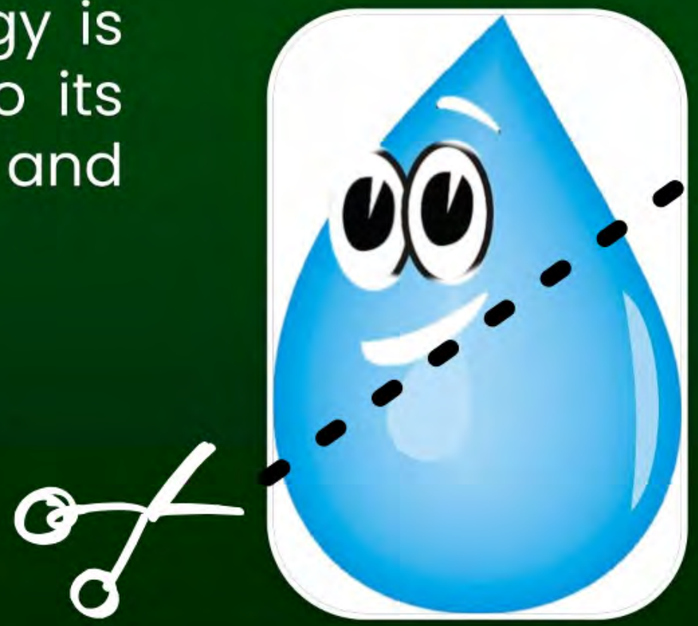
- **Step I. Activation of chlorophyll:** The chlorophyll on exposure to light energy becomes activated by absorbing photons (photon is the smallest unit of light energy).

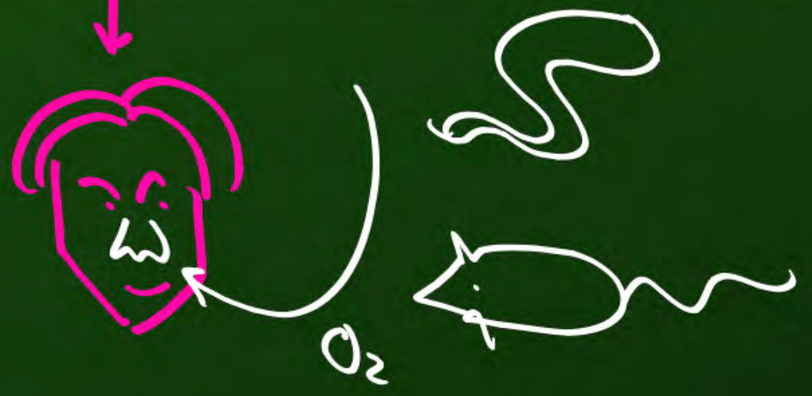


- **Step II. Splitting of water:** The absorbed energy is used in splitting the water molecule (H_2O) into its two components (Hydrogen and Oxygen) and releasing electrons.

$$2H_2O \xrightarrow{\text{energy of 4 photons}} 4H^+ + 4e^- + O_2\uparrow$$

Photolysis





- The reaction is known as photolysis, which means splitting by light (photo = light, lysis = breaking).
- Photolysis occurs in the grana of a chloroplast and is defined as the splitting of H_2O molecules into hydrogen ions and oxygen in the presence of light.

$$NADP^+ + e^- + H^+ \xrightarrow{\text{enzyme}} NADPH$$

End Products of Photolysis

(1) The hydrogen ions (H^+) are picked up by a compound NADP (Nicotinamide adenine dinucleotide phosphate) to form NADPH $NADP^+ + e^- + H^+ \xrightarrow{\text{Enzyme}} NADPH$	**(2)** The oxygen (O) component is given out as molecular oxygen (O_2) $2O \rightarrow O_2$ This oxygen is used in respiration for leaf called as photorespiration	**(3)** The electrons (e^-)are used in converting ADP into energy-rich compound ATP by adding one phosphate group Pi (inorganic phosphate). ADP + Pi (inorganic phosphate) $\rightarrow$ ATP This process is called phosphorylation or photophosphorylation.

Light-independent phase (Biosynthetic Phase)

Dark Reaction

- The dark reactions use the ATP and NADPH produced during the light reactions for energy
- Does not depend on light
- Occur in stroma of chloroplast
- Carbon dioxide is converted into Glucose.
- Also known as Calvin cycle

$$CO_2 \xrightarrow[NADPH]{ATP} C_6H_{12}O_6$$

QUESTION

Which one of the following is NOT a requirement for photosynthesis?

1. Chlorophyll
2. CO_2
3. O_2
4. More than one of the above

By product



Thank You